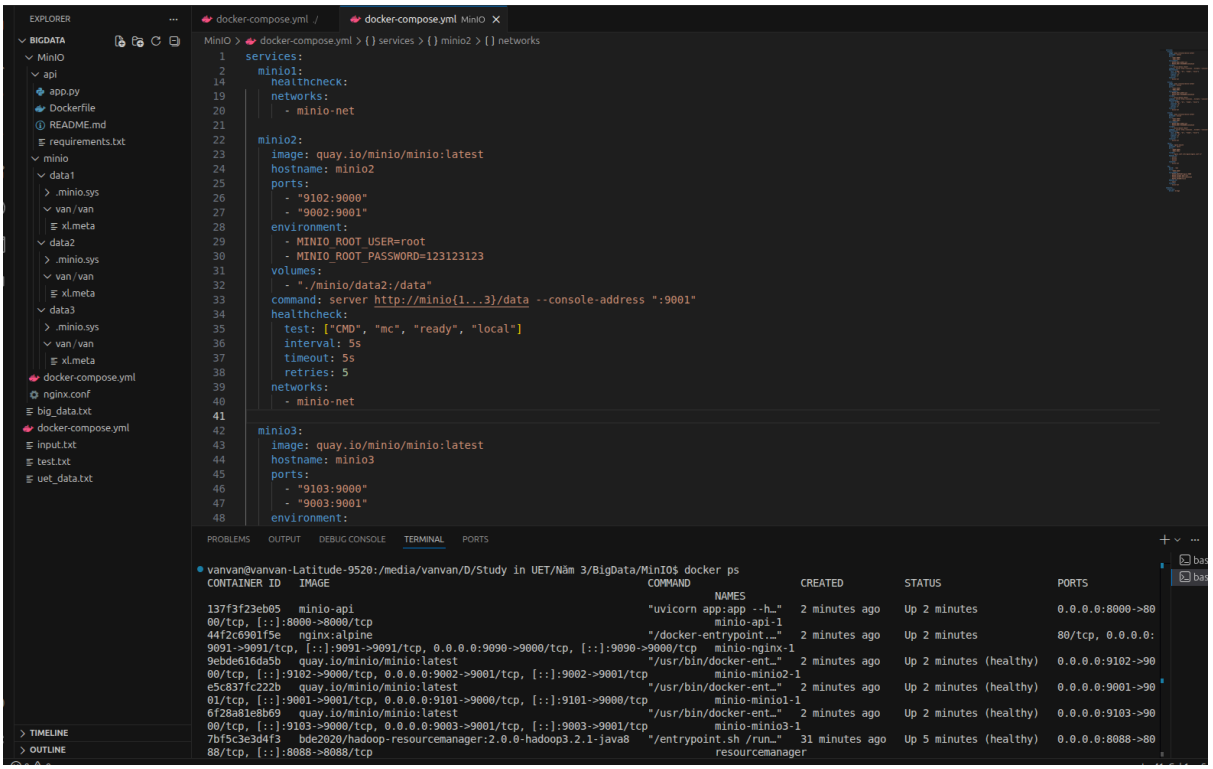
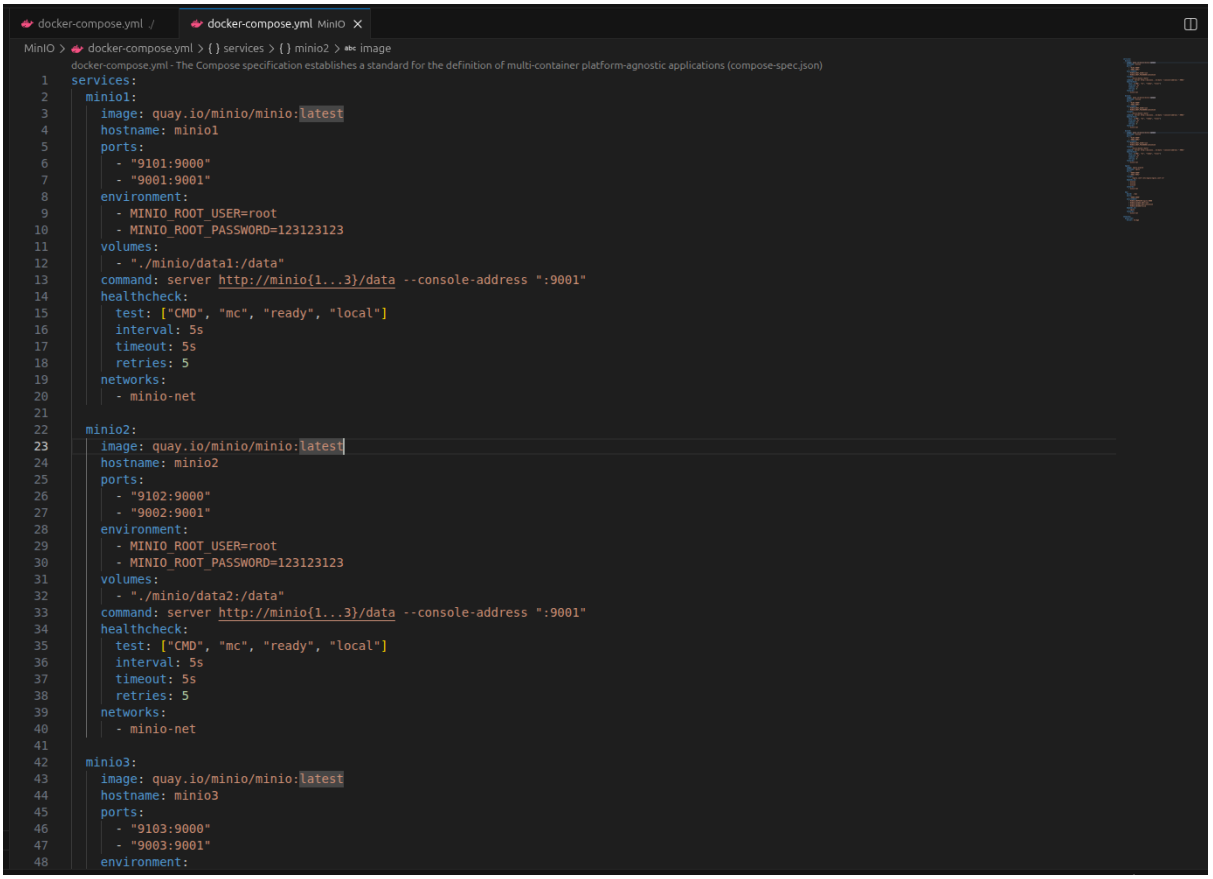
1. Cài đặt cụm lưu trữ Object Storage gồm 3 máy





docker-compose.yml ./ | docker-compose.yml MinIO | app.py U

MinIO > api > app.py > { } FastAPI

```python
from fastapi import FastAPI, UploadFile, File, HTTPException, Query
from fastapi.responses import StreamingResponse
from minio import Minio
from minio.error import S3Error
import io
import os

MINIO_ENDPOINT = os.getenv("MINIO_ENDPOINT", "localhost:9090")
MINIO_ACCESS_KEY = os.getenv("MINIO_ACCESS_KEY", "root")
MINIO_SECRET_KEY = os.getenv("MINIO_SECRET_KEY", "123123123")
MINIO_SECURE = os.getenv("MINIO_SECURE", "false").lower() == "true"

client = Minio(
    MINIO_ENDPOINT,
    access_key=MINIO_ACCESS_KEY,
    secret_key=MINIO_SECRET_KEY,
    secure=MINIO_SECURE,
)

app = FastAPI(title="MinIO API", version="1.0.0")


# ── Buckets ──────────────────────────────────────────────────────────

@app.get("/buckets", tags=["Buckets"])
def list_buckets():
    buckets = client.list_buckets()
    return [{"name": b.name, "created_at": b.creation_date} for b in buckets]


@app.post("/buckets/{bucket_name}", tags=["Buckets"])
def create_bucket(bucket_name: str):
    if client.bucket_exists(bucket_name):
        raise HTTPException(status_code=409, detail=f"Bucket '{bucket_name}' already exists")
    client.make_bucket(bucket_name)
    return {"message": f"Bucket '{bucket_name}' created"}


@app.delete("/buckets/{bucket_name}", tags=["Buckets"])
def delete_bucket(bucket_name: str):
    if not client.bucket_exists(bucket_name):
        raise HTTPException(status_code=404, detail=f"Bucket '{bucket_name}' not found")
    try:
        client.remove_bucket(bucket_name)
    except S3Error as e:
        raise HTTPException(status_code=400, detail=str(e))
    return {"message": f"Bucket '{bucket_name}' deleted"}


```

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MinIO > api > app.py > { } FastAPI

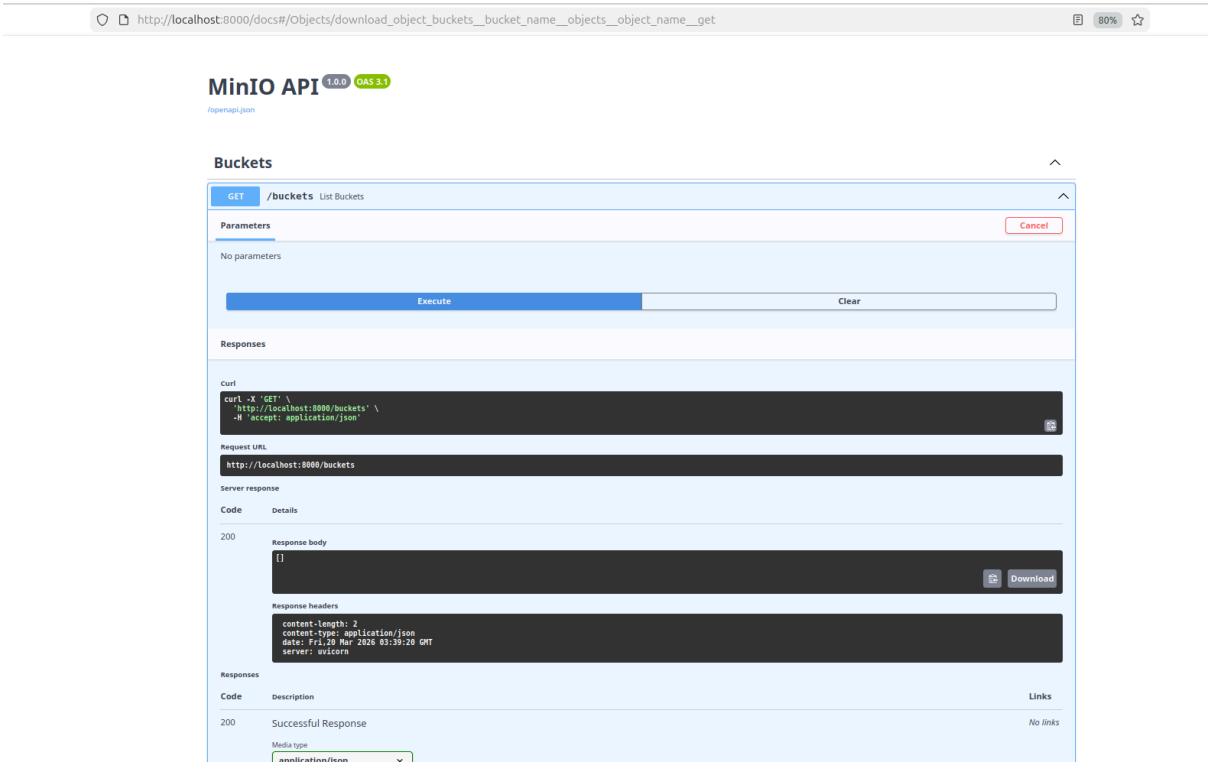
```python


# ── Objects ──────────────────────────────────────────────────────────

@app.get("/buckets/{bucket_name}/objects", tags=["Objects"])
def list_objects(
    bucket_name: str,
    prefix: str = Query(default="", description="Lọc theo prefix"),
    recursive: bool = Query(default=True),
):
    if not client.bucket_exists(bucket_name):
        raise HTTPException(status_code=404, detail=f"Bucket '{bucket_name}' not found")
    objects = client.list_objects(bucket_name, prefix=prefix, recursive=recursive)
    return [
        {
            "name": obj.object_name,
            "size": obj.size,
            "last_modified": obj.last_modified,
            "etag": obj.etag,
        }
        for obj in objects
    ]


@app.post("/buckets/{bucket_name}/objects", tags=["Objects"])
async def upload_object(
    bucket_name: str,
    file: UploadFile = File(...),
    object_name: str = Query(default=None, description="Tên lưu trên MinIO, mặc định là tên file"),
):
    if not client.bucket_exists(bucket_name):
        raise HTTPException(status_code=404, detail=f"Bucket '{bucket_name}' not found")

    name = object_name or file.filename
    data = await file.read()

    client.put_object(
        bucket_name,
        name,
        io.BytesIO(data),
        length=len(data),
        content_type=file.content_type or "application/octet-stream",
    )
    return {"message": f"Uploaded '{name}' to bucket '{bucket_name}'", "size": len(data)}


@app.get("/buckets/{bucket_name}/objects/{object_name:path}", tags=["Objects"])
def download_object(bucket_name: str, object_name: str):
    if not client.bucket_exists(bucket_name):
```

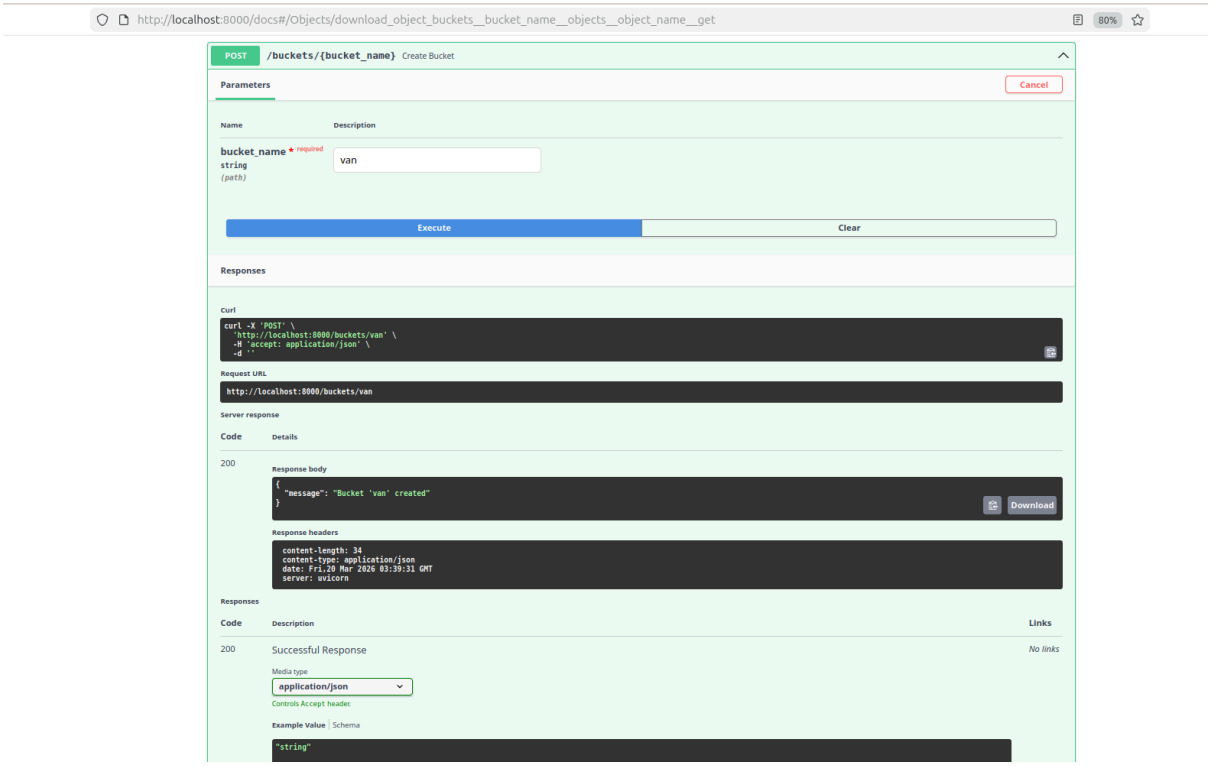
docker-compose.yml ./ | docker-compose.yml MinIO | app.py U

MinIO > api > app.py > { } FastAPI

```python
def download_object(bucket_name: str, object_name: str):
    except S3Error as e:
        raise HTTPException(status_code=404, detail=str(e))


@app.delete("/buckets/{bucket_name}/objects/{object_name:path}", tags=["Objects"])
def delete_object(bucket_name: str, object_name: str):
    if not client.bucket_exists(bucket_name):
        raise HTTPException(status_code=404, detail=f"Bucket '{bucket_name}' not found")
    try:
        client.remove_object(bucket_name, object_name)
    except S3Error as e:
        raise HTTPException(status_code=400, detail=str(e))
    return {"message": f"Deleted '{object_name}' from bucket '{bucket_name}'"}


# ── Presigned URLs ──────────────────────────────────────────────────────

@app.get("/buckets/{bucket_name}/presign/{object_name:path}", tags=["Presigned"])
def presign_download(
    bucket_name: str,
    object_name: str,
    expires: int = Query(default=3600, description="Thời gian hết hạn (giây)"),
):
    from datetime import timedelta
    try:
        url = client.presigned_get_object(
            bucket_name, object_name, expires=timedelta(seconds=expires)
        )
        return {"url": url, "expires_in": expires}
    except S3Error as e:
        raise HTTPException(status_code=400, detail=str(e))


@app.post("/buckets/{bucket_name}/presign/{object_name:path}", tags=["Presigned"])
def presign_upload(
    bucket_name: str,
    object_name: str,
    expires: int = Query(default=3600, description="Thời gian hết hạn (giây)"),
):
    from datetime import timedelta
    try:
        url = client.presigned_put_object(
            bucket_name, object_name, expires=timedelta(seconds=expires)
        )
        return {"url": url, "expires_in": expires}
    except S3Error as e:
        raise HTTPException(status_code=400, detail=str(e))
```

2. Sử dụng S3 API thực hiện các REST action trên cụm dữ liệu





http://localhost:8000/docs#/Objects/download_object_buckets__bucket_name__objects__object_name__get

GET /buckets/{bucket_name}/objects List Objects

Parameters

Cancel

Name	Description
bucket_name * required string (path)	van
prefix string (query)	Lọc theo prefix prefix
recursive boolean (query)	true

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8000/buckets/van/objects?recursive=true' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8000/buckets/van/objects?recursive=true

Server response

Code	Details
200	Response body [] Download Response headers content-length: 2 content-type: application/json date: Fri,20 Mar 2026 03:39:48 GMT server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

http://localhost:8000/docs#/Objects/download_object_buckets__bucket_name__objects__object_name__get

POST /buckets/{bucket_name}/objects Upload Object

Parameters

Cancel Reset

Name	Description
bucket_name * required string (path)	van
object_name string (query)	Tên lưu trên MinIO, mặc định là tên file van

Request body required multipart/form-data

file * required
string
Browse... Screenshot_20260311_172017.png

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/buckets/van/objects?object_name=van' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@Screenshot_20260311_172017.png;type=image/png'
```

Request URL

http://localhost:8000/buckets/van/objects?object_name=van

Server response

Code	Details

200	Response body
	{
	"message": "Uploaded 'van' to bucket 'van'",
	"size": 238846
	}
	Download
	Response headers
	content-length: 58
	content-type: application/json
	date: Fri,20 Mar 2026 03:40:14 GMT
	server: uvicorn

http://localhost:8000/docs#/Objects/download_object_buckets__bucket_name__objects__object_name__get

GET /buckets/{bucket_name}/objects/{object_name} Download Object

Parameters

Cancel

Name	Description
bucket_name * required string (path)	van
object_name * required string (path)	van

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8000/buckets/van/objects/van' \
  -H 'accept: application/json'
```

Request URL

```
http://localhost:8000/buckets/van/objects/van
```

Server response

Code	Details
200	Response body Download file Response headers

```
content-disposition: attachment; filename="van"
content-type: application/octet-stream
date: Fri,20 Mar 2026 03:40:25 GMT
server: uvicorn
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value | Schema	No links

http://localhost:8000/docs#/Objects/delete_object_buckets__bucket_name__objects__object_name__delete

DELETE /buckets/{bucket_name}/objects/{object_name} Delete Object

Parameters

Cancel

Name	Description
bucket_name * required string (path)	van
object_name * required string (path)	van

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:8000/buckets/van/objects/van' \
  -H 'accept: application/json'
```

Request URL

```
http://localhost:8000/buckets/van/objects/van
```

Server response

Code	Details
200	Response body Download Response headers

```
{
  "message": "Deleted 'van' from bucket 'van'"
}
```

```
content-length: 45
content-type: application/json
date: Fri,20 Mar 2026 03:49:12 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response Media type application/json	No links

http://localhost:8000/docs#/Buckets/delete_bucket_buckets__bucket_name__delete

80%

DELETE `/buckets/{bucket_name}` Delete Bucket

Parameters Cancel

Name	Description
bucket_name * required string *(path)*	van

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:8000/buckets/van' \
  -H 'accept: application/json'
```

Request URL

```
http://localhost:8000/buckets/van
```

Server response

Code	Details

200	Response body
	`{ "message": "Bucket 'van' deleted" }`
	Download
	Response headers
	`content-length: 34`
	`content-type: application/json`
	`date: Fri,20 Mar 2026 03:49:47 GMT`
	`server: uvicorn`

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value | Schema `"string"`	*No links*